INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

Manufacturing

The complete set — what is in it, what each piece is for, and the order to pick them up

Two volumes, six spreadsheets, three decks and the worked example they were all built against. Read this first: it says which file answers which question, and which three you need in your first week.

South African lens. Sourced throughout. October 2026.

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Contents

THE SET

Guide 1 · What is in this set

GUIDE 1

What is in this set

Fifteen files. Three of them matter in your first week.

This set exists so that an architect who has never worked in a factory can walk into one and be useful. Not eventually — within a quarter.

It comes in four parts and a worked example. The two books teach and instruct, the spreadsheets are what you actually fill in, the decks are what you put on a screen in front of other people, and the Kestrel files are a complete, populated example of what the finished thing looks like.

Read these three first

> **IF YOU READ NOTHING ELSE**
>
> **`01 Learner guide`** — read it once, front to back, before you start. It is the industry.
>
> **`02 Workbook`** — never read front to back. Open it at the phase you are in and work the numbered steps.
>
> **`03 Excel templates / Manufacturing_EA_Metamodel_TEMPLATE.xlsx`** — the empty repository. Everything the workbook tells you to do lands in it.

The whole set

Folder	File	What it is	When you use it
01	Manufacturing — a starter pack for enterprise architects	The learner guide. 234 pages, 23 items.	Read once, keep beside you
02	Manufacturing — the architecture workbook	The ADM workbook. 140 numbered steps, 30 printable forms.	Every working day
03	Manufacturing_EA_Metamodel_TEMPLATE.xlsx	The empty repository. 119 sheets, with the change control and the eight sheets the example lacks.	From day one
03	Manufacturing_Architecture_Comparison_Tool_TEMPLATE.xlsx	Baseline against target, with the governed-drift test.	Phase E, then quarterly
03	Manufacturing_EA_Workshop_Capture.xlsx	Live capture for the five workshops.	In the room
03	Manufacturing_EA_Transcription_Forms.xlsx	30 forms matching the printed ones.	Typing up paper
03	Manufacturing_EA_Principles_and_Standards.xlsx	48 live principles, 31 standards, ready to adopt or argue with.	Preliminary phase
03	Manufacturing_EA_Maturity_Assessment.xlsx	26 scored dimensions with the evidence for each.	First month, then annually
04	Manufacturing_EA_Executive_Briefing.pptx	31 slides for the executive committee.	End of the first quarter
04	Manufacturing_EA_View_Templates.pptx	49 slides: all 40 ArchiMate 4 views, empty and guided.	Whenever you draw
04	Manufacturing_EA_Workshop_Pack.pptx	50 slides facilitating the five workshops.	In the room
05	Manufacturing_TOGAF_Metamodel.xlsx	The worked example — a populated repository for an invented manufacturer.	When you want to see what finished looks like
05	Manufacturing_ArchiMate4_Views.pptx	Worked ArchiMate 4 views of that repository.	Alongside the view templates

The two books, and why they are separate

	Volume 1 — the learner guide	Volume 2 — the workbook
Read	once, front to back	never front to back
Answers	what this industry is and how it runs	what do I do on Monday
Shelf life	keep it as reference	consumed — you write in it

Volume 1 covers the archetypes, the vocabulary, the business model, who does what, how manufacturers make money, the operating model and the planning cadence, the process model in BPMN, the data and information models, the application landscape, the plant floor, the reference architectures such as they are, the providers, the regulator and the inspector, and how to model all of it in ArchiMate 4. Volume 2 takes the ADM phase by phase and says which sheet, which columns, who signs it, how you know it is done, and how it usually goes wrong.

They interlock. Every step in volume 2 with teaching behind it carries a **Look it up** line naming the volume 1 chapter and section, and volume 2's appendix E indexes all 140 steps back to volume 1.

The chapter no other volume in this series has

Volume 1 chapter 13 is **the plant floor — OT, ISA-95 and the IT/OT boundary**. Banking has no equivalent. Nor does insurance or retail. Below the ERP a manufacturer runs a second estate — PLCs, a DCS, SCADA servers, a historian, a plant network — and it obeys different rules: you cannot patch it on a Tuesday, it must keep producing when the corporate network is down, and the number it reports is not always the number it recorded.

That chapter is the reason this set is longer than its siblings, and it is the chapter to read twice. Volume 2's Phase D chapter is its working half.

Print volume 2

Volume 2 is built to be printed and written in: 30 printable forms whose columns are the real columns of the real sheets, 64 question sets with ruled answer space, and working tables inside the steps.

> **THE TRANSCRIPTION LOOP**
>
> Paper in the room, spreadsheet the same afternoon, repository record with a change reference. Paper that is never typed up is a wasted workshop; a record typed up without a change reference is a repository that will rot inside a year.

The notation

Everything that is architecture is modelled in **ArchiMate 4** — 42 elements, 11 relationships. The one exception is level-2 process models, drawn in BPMN 2.0, because ArchiMate has no swimlanes and in

a process model the hand-off is the point. Volume 1 chapter 17 teaches the notation; volume 2 appendix D maps all 64 object sheets and 25 relationship sheets of the repository onto it — and says plainly that **26 of the 64 have no clean equivalent**, that **fifteen of the 25 relationships reverse**, and that two cannot be expressed at all.

One of those two is worth knowing before you start: **there is no metric element in ArchiMate, and `Value` is not a number.** So OEE, yield and on-time-in-full have no home in the notation. They live in a measure definition register, which is a separate artefact, and which this set tells you to build before you build a dashboard.

Manufacturing has no reference model, and that changes how you use this set

Banking has BIAN. Insurance has ACORD. Manufacturing has a control-systems hierarchy standard, a batch-control standard, three competing interoperability protocol families from three different bodies with no reconciliation between them, an identification system, a handful of engineering-exchange formats, and a stack of sector quality schemes. They are maintained by different organisations and they do not cover the same ground. There is no vendor-neutral industry model you can adopt and populate.

That is not a gap in this set — it is the industry's condition, and volume 1 chapter 14 says so with evidence. The practical consequence is that you build the house reference model yourself, from four sourced inputs: ISA-95 object names for the ERP/MES boundary, the applicable quality scheme's record obligations for the data-ownership matrix, ISO 22400 for measure definitions, and ISA/IEC 62443 zones and conduits for technology and security — and you declare in your principles that no single industry reference model governs.

The honest counter-indication is in the book too. Catena-X is the nearest thing to a maintained, multi-company, vendor-neutral manufacturing data architecture, and it is the single most likely way this material dates.

What the worked example gets wrong, on purpose and not

The Kestrel repository is a teaching artefact and it has real defects. The books do not smooth them over — they are the best material in the set, because every one of them is a defect you will meet in a real repository.

Its index reports 843 records; the sheets hold 813, and the difference is exactly the thirty interface rows on one relationship sheet. It claims every object type carries records while the compliance-assessment sheet is empty — so the design gate has produced nothing, which is also what the one absent security control says. Two sheets both claim to name the golden source for a data entity and **disagree on nine of the ten they share**, and the item master is recorded as mastered in the historian, which is an append-only time-series store and cannot master anything. The supplier, the quality specification and the inventory position are all classified Public. Thirteen relationship sheets pair row *n* of one list with row *n* of another, which is how ISO 9001 comes to be evidenced by identity lifecycle management. A report asserts four platform components are past vendor support; two are, and the other two have

support ends in 2027, because the column was typed rather than computed. The regulatory sheet is for a European manufacturer — CE and UKCA marking, which have no application in South Africa.

And the sharpest one: **there is no factory in it.** The actor and role sheets hold the same eighteen corporate titles — chief executive down to data steward — and there is no plant manager, no production planner, no scheduler, no maintenance planner, no quality manager, no process engineer, no operator and no millwright anywhere in 813 records. The production line, which is the object OEE is measured on, is declared in the industry extension and exists nowhere else in the file.

The template in folder 03 ships with those gaps fixed and says on its face what it changed.

Two invented manufacturers, kept apart

Karoo Industrial Group is volume 1's running example — a composite with a batch food business and a discrete automotive components plant, deliberately two archetypes under one roof, because almost every real plant is several at once. Chapter 18 walks it end to end. **Kestrel Industrial Group** is the manufacturer inside the worked-example repository. Both are inventions, and every figure attributed to either is illustrative. The worked example keeps its money in dollars; a South African reader should read the magnitudes, not the currency.

What this set does not do

It does not pick your vendor — the providers chapter names who plays in each category and says plainly, at both ends, that these are examples and not recommendations. It does not replace legal advice: the regulatory chapters tell you what the inspector asks to see and which columns are the evidence, not whether your arrangement complies. It quotes no benchmark it could not source, which is why you will not find the 85 per cent "world-class" OEE figure anywhere in it except as a warning. And it does not fill itself in, because a repository somebody else populated is a document, not an architecture.